FYP-2 Mid Evaluation — Presenter Division and Speaking Scripts

Project S26-017-R-GASR — 4D Surgical Scene Reconstruction and Deformation Prediction

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Deck: **FYP2_Mid_Eval_Modules_5_and_6.pptx** — 16 slides · Total speaking time ≈ 18 minutes

1. How the Slides Are Divided

The split follows the subsystem each member owns in the codebase, recorded in *Module5_6_Individual_Contributions.docx*. Each member presents the slides whose numbers their own code produced. Nobody presents a result they did not measure.

Member	Slides	Time	What they own on stage
Ahmad Rashad 22i-1175	1, 2, 5, 6, 7, 8 (6 slides)	≈ 6 min 20 s	Opens the talk. The problem, both modules at a conceptual level, the architecture of both, and every problem and fix inside Module 5 — the network, the diffusion formulation and the training system are his.
Shehroz 22i-0771	9, 10, 11, 12, 13 (5 slides)	≈ 6 min 40 s	The entire evidence block. Module 6's measurement problems, the training outcome, the baseline comparison, the key diagnostic finding and the uncertainty result. Every number on these five slides came out of his evaluation framework.
Humair Khan 22i-2623	3, 4, 14, 15, 16 (5 slides)	≈ 5 min 20 s	Bookends the talk with where the field is and where we sit in it: the FYP-1 reconstruction he helped render and export, the literature positioning, the comparison against published work, what comes next, and the close.

Why this division and not an even 5-5-6 cut by slide number

Three reasons, and the panel is likely to ask about at least one of them:

- **Ahmad takes slides 5, 6 and 7 together** because those three are descriptive — what each module is and how they connect. He designed the interface between them (a checkpoint file and an

architecture descriptor), so he is the right person to explain how the two halves meet. On slide 7 he explicitly names Shehroz as the owner of the lower band.

- **Shehroz takes slide 10** (the training result) even though Ahmad built the training system. The slide reports a *measured* held-out number, and held-out measurement is Shehroz's subsystem. He credits Ahmad's training system in one line.
- **Humair takes slide 4** (literature) and **slide 14** (comparison with published work) because both are about positioning against the reconstruction literature, which is where his rendering and export work sits, and the reconstruction PSNR on slide 14 is produced by the renderer he re-implemented and verified.

Handover map — four handovers in total

After slide	From → To	Cue line
2	Ahmad → Humair	"Before we get into the two modules, Humair will show you what FYP-1 already delivered, and where this sits in the literature."
4	Humair → Ahmad	"That is the gap. Ahmad will now take you through the two modules that go after it."
8	Ahmad → Shehroz	"Those are Module 5's problems. Shehroz built everything that measures whether the fixes worked, so he will take the rest of the evidence."
13	Shehroz → Humair	"That is where our own numbers stand. Humair will close by putting them next to the published work."

> **Practical note.** Four handovers means four pauses. Agree beforehand that the person handing over takes one step back from the laptop and the next person steps in — do not pass the clicker mid-sentence. If you are presenting over a shared screen instead, agree who drives the deck for the whole talk and the others just say "next slide, please."

2. Ahmad Rashad (22i-1175) — Slides 1, 2, 5, 6, 7, 8

Opening, the problem, both modules, the architecture, and Module 5's problems and fixes. ≈ 6 minutes 20 seconds.

Slide 1 — Title · 30 seconds

On screen: "From Reconstruction to Anticipation", subtitle, the three names, supervisor, department.

Good morning. I am Ahmad Rashad, and with me are Shehroz and Humair. Our supervisor is Dr Akhtar Jamil.

This is our FYP-2 mid evaluation. In FYP-1 we built a system that reconstructs a surgical scene in 3D from a single laparoscopic camera. That work is complete. FYP-2 asks a harder question: can the same system tell you what the tissue is about to do *next*? That is Modules 5 and 6, and that is what this presentation is about.

We will be direct with you about the result, because the interesting part of this project is not a number we beat — it is what we found out when we did not.

Slide 2 — The laparoscope shows the present, never the next moment · 1 minute

On screen: left text column, two cards (What exists today / What is missing), four stat tiles.

Here is the clinical situation. A surgeon operates through one camera. That camera gives a flat image. There is no depth, no 3D structure, and crucially no indication of how the tissue will respond to the instrument that is about to touch it.

Reconstruction research has solved the first half of that problem very well. It has not addressed the second half at all. Every published surgical 4D reconstruction system is retrospective — it models deformation that has *already* been recorded.

Point at the stat tiles.

And these four numbers are why this is hard. We have zero depth sensors — it is a single monocular camera. We have zero deformation labels, because no surgeon annotates tissue motion frame by frame. We have ten reconstructed videos, which is a small corpus. And the thing we are trying to predict moves one to two pixels per frame. That last number is the real difficulty: the signal we are chasing is smaller than the noise in most of what we are measuring.

Handover: "Before we get into the two modules, Humair will show you what FYP-1 already delivered, and where this sits in the literature."

Slide 5 — Module 5: learning how tissue moves · 1 minute 20 seconds

On screen: Input / Model / Output cards, two gold cards, mint strip at the bottom.

Thank you, Humair. So that is the gap. Module 5 is our attempt to close it.

Module 5 is a conditional diffusion model. It learns the deformation of 512 tracked tissue points, and it is trained entirely without clinical labels.

Walk the three cards left to right, roughly ten seconds each.

The input is three things: the last four frames of those 512 points, a number saying how far ahead we want to predict — one, two or three frames — and a patch of random noise. The model is 16.3 million parameters: a temporal transformer that looks along time for each point separately, then two-way cross-

attention, and it denoises over twenty steps. The output is the predicted movement of each point — and because we can re-run it with fresh noise, we get several plausible futures from the same input rather than one guess.

Now the two gold cards. These are the two decisions the panel is most likely to challenge, so take them deliberately.

Two design decisions worth defending. First, why diffusion rather than direct regression. Soft tissue under contact is genuinely uncertain — there is no single correct future. A single regression estimate hides that. Sampling the model repeatedly gives us a spread of outcomes, and Shehroz will show you later that this spread is a usable confidence signal.

Second, why the horizon is an input. In our original formulation, the same four frames appeared in training with targets one, two and three frames ahead — three different right answers for identical input. Without telling the model which one we wanted, the task is unlearnable no matter how large the network is. Embedding the horizon and injecting it into both branches is what made the objective well posed.

All of it trained on 7,371 windows across the ten videos, with no human annotation at any stage.

Slide 6 — Module 6: using the model, and measuring it honestly · 1 minute

On screen: three teal cards, two brick cards, mint strip.

Module 6 trains nothing. It loads the trained checkpoint, asks it three different questions, and grades every answer.

Ask once: where will the tissue be one, two or three frames from now? That is the most reliable mode. Ask repeatedly: feed each answer back in and reach ten frames ahead — error compounds, and we measure how fast. Ask five times with different noise: five possible futures from one input, and how much they disagree becomes the confidence estimate.

The brick card on the left is the one to emphasise.

And everything is graded against two deliberately stupid rules. "Nothing moves" — the naive baseline. And "everything continues in a straight line" — constant velocity. Our position is simple: if a 16 million parameter model cannot beat those two, we say so. We do not change the metric until it looks better.

Six quality tests sit on top of that. The first one, QT0, verifies that the data we are evaluating on is genuinely the data the model trained on, before anything is predicted at all. I will say more about why that matters when Shehroz gets to it.

Slide 7 — Architecture of both modules · 1 minute 10 seconds

On screen: two bands. Top = Module 5 training. Bottom = Module 6 inference and evaluation. The vertical arrow on the right is the checkpoint.

This is both modules on one slide, and the important thing is the arrow on the right.

Trace the top band left to right with your hand.

The top band is Module 5, which is mine. Four frames of 512 points go into a shared point encoder, then a temporal transformer — three layers, eight heads — then cross-attention in both directions, then an output head that predicts velocity. The yellow box underneath is the conditioning: the diffusion step and the prediction horizon are injected into the transformer. That horizon arrow is the fix I described two slides ago, and it is the single change that made the problem well posed.

The band emits one thing: a trained checkpoint.

Follow the vertical arrow down to the lower band.

The bottom band is Module 6, which is Shehroz's and Humair's. It consumes that checkpoint and never trains. It replays the trajectory with a verification gate, runs the DDIM sampler for twenty steps, and branches into the three question modes. All three are compared against the same two baselines, and the whole thing exports metrics and figures for the comparative study.

The interface between the two halves is deliberately narrow — one checkpoint file and an architecture descriptor — which is what let the three of us work in parallel without blocking each other.

Slide 8 — Module 5: what went wrong, and what we did · 1 minute 20 seconds

On screen: five problem/solution rows. Do not read all ten boxes aloud — summarise the first three quickly and spend your time on the last one.

Five problems in Module 5, each one found by a measurement rather than a guess.

The task had no single right answer — fixed by conditioning on the horizon, which I have already described. The signal was smaller than the noise: tissue moves 0.004 units while diffusion adds noise at scale 1, so we rescale the target to unit variance and invert that before every reported metric, so every number you see is in physical units. The raw trajectories had almost no momentum — correlation between consecutive motions was 0.077, essentially noise — so we added per-video adaptive smoothing and that correlation rose to 0.89.

Fourth, training made the model worse after epoch 60. Held-out error rose while training loss kept falling. We measured that rather than assumed it, shortened the schedule and raised weight decay, and we keep the best checkpoint, not the last one.

Slow down here. This is the most important design change in the project.

And the fifth is the one I want you to remember. We found that a model which simply gave up — predicted zero displacement everywhere — scored exactly the same as doing nothing. The naive baseline *was* the degenerate solution. So we changed what the model predicts: instead of predicting the displacement, it now predicts the *correction* to constant-velocity extrapolation. That means if it gives up now, it falls back to an estimator that is already 41% better than naive. We moved the floor.

Handover: "Those are Module 5's problems. Shehroz built everything that measures whether the fixes

worked, so he will take the rest of the evidence."

Ahmad — questions likely to come to you

- **"Why diffusion? Isn't this overkill for 512 points?"** — Two reasons. The uncertainty is real and clinically meaningful, and diffusion gives it to us for free by sampling. And slide 13 shows that spread correlates with actual error at 0.69, which no deterministic model can produce. The parameter count is driven by the transformer, not the diffusion.
- **"Why 20 sampling steps and not 500?"** — Training defines 500; DDIM lets us sample in 20 with negligible loss. Module 8 studies that trade-off explicitly.
- **"You trained on 10 videos. Isn't that far too few?"** — Yes, and we say so on slide 15. The overfitting we measured after epoch 60 is exactly what you would expect at this corpus size. It is the fifth item on our future-work list for that reason.
- **"Why predict velocity rather than noise?"** — At a displacement scale of 0.004, noise prediction is numerically badly conditioned. The v-parameterisation stays well conditioned across the whole schedule.

3. Shehroz (22i-0771) — Slides 9, 10, 11, 12, 13

The evidence block: Module 6's measurement problems, the training outcome, the comparison, the key finding and the uncertainty result. ≈ 6 minutes 40 seconds.

Shehroz — the tone to hold. Your five slides contain the one result that does not flatter us. Do not apologise for it and do not bury it. Report it the way you measured it, then show that you found the cause. A panel trusts a team that reports a loss accurately far more than one that explains it away. Slides 12 and 13 are where you win this presentation back.

Slide 9 — Module 6: what went wrong, and what we did · 1 minute 10 seconds

On screen: five problem/solution rows. Lead with the first; it is the one that shows methodological seriousness.

Thank you, Ahmad. Module 6's problems are different in kind from Module 5's. They are almost all about measurement integrity rather than modelling, which I think is the right emphasis for an evaluation module.

The first one is the one I care most about. Rollout evaluation needs continuous trajectories, but Module 4 stores fixed windows. So we rebuild each video's trajectory by replaying the upstream processing — same random seed, same smoothing window, same normalisation. But a rebuild can silently diverge, and if it did, every number in this project would be wrong and we would never know. So QT0 cross-checks the rebuild against the windows Module 4 actually saved, and it runs as a *gate* — nothing is predicted until it

passes.

Second, the metric was penalising us unfairly. Chamfer Distance is minimised by the average of the distribution, not by any single sample, so scoring one sample charged the model a penalty of roughly 40% for free. We now average several sampled trajectories before scoring.

Third, our diagnostics were measuring one easy video — taking the first N windows drew almost entirely from a single recording. Spreading the evaluation set evenly across all ten showed that our earlier scores had been about four times too optimistic. We would rather know that.

The last two are Humair's. Nothing on screen was actually a prediction — every predicted frame was paired with a frame that already existed — so he made the forecast run twelve frames past the end of the recorded video. And predicted motion was invisible at one to two pixels, so he added magnified motion arrows, with the magnification factor printed on every panel and never applied to any number.

Slide 10 — Training converged cleanly, and then overfitted · 1 minute 10 seconds

On screen: four stat tiles, two cards, one wide statement box at the bottom.

So what came out of training. Ahmad's training system ran 500 epochs on 16.3 million parameters. The training loss fell from 0.36 to 0.012, and the best held-out checkpoint was at about epoch 60.

What went right: smooth convergence, no instability, no divergence. And because Colab disconnects, multi-session training resumes exactly where it stopped — that mattered more than it sounds.

What went wrong: held-out error bottomed out near epoch 60 and rose after about 260 while training loss kept falling. That is overfitting, and we measured it directly rather than assuming it.

Pause before the bottom box. Let them read it.

And here is the number. Our best held-out Chamfer Distance is 0.01568. The naive baseline — predicting that nothing moves at all — is 0.01557. We are very slightly worse than doing nothing.

The model converged to predicting almost no motion, which is exactly what the naive baseline does. Understanding *why* is the rest of this presentation.

Slide 11 — A two-line formula beats our model · 1 minute 20 seconds

On screen: the comparison table, the brick card on the right, the wide box at the bottom.

This is the full comparison, on the complete held-out set across all ten videos.

Walk the table rows, not the individual cells.

Nothing moves, constant velocity, and our diffusion model, at one, two and three frames ahead. Read it honestly: our model sits essentially exactly on the naive baseline at every horizon. Constant velocity — which is two lines of code and has no parameters at all — beats both of us, by 41% at one frame ahead. And we lose on all ten videos, not on some subset we could explain away.

Now the bottom box. This is where you show the work.

Before concluding anything, we tested two explanations that would have let us off the hook, and eliminated both.

The first: maybe error grows the further we get from the training frames — drift. We correlated error against distance from training data and got rho of plus 0.02 with a p-value of 0.61. There is no drift.

The second: maybe dropped frames during preprocessing corrupted the timing. We checked all 333 evaluation origins and found zero dropped frames.

Rollout is also stable — error grows with an exponent of 0.61, which is sub-linear, so the model is not diverging. Eliminating those two explanations by measurement is what let us find the actual cause, which is the next slide.

Slide 12 — The model knows which way tissue moves, not how far · 1 minute 40 seconds

On screen: a large teal statement, five cosine tiles, two cards. This is the most important slide in the deck — spend the time.

Chamfer Distance alone cannot tell the difference between a model that learned nothing and a model that learned the right direction but the wrong distance. Both score the same. So we measured them separately.

These five numbers are the agreement between the direction we predict and the direction the tissue actually moves — cosine similarity, where zero means no relationship at all.

Point at each tile as you say it.

At one frame ahead, plus 0.57. At two, plus 0.40. At three, plus 0.26. It decays, and by five frames it is at zero and by ten it is negative — but at the horizons the model was actually trained for, it is strongly positive.

Let that land before the interpretation.

Here is why that matters. A model that had genuinely learned nothing would read 0.00 everywhere. It does not. Plus 0.57 combined with naive-level error can only mean one thing: the direction is right and the magnitude is wrong. And that is a far more tractable problem than "the model does not work."

Two fixes follow directly from that measurement, and both are already implemented. First, predict the correction to constant velocity rather than the displacement itself — Ahmad described that on slide 8 — which moves the fallback from the naive baseline to an estimator 41% stronger. Second, calibrate the magnitude at inference: we solve for a single scale factor that best matches predicted motion to observed motion. The mathematics there is worth one sentence — once you scale optimally, the remaining error is proportional to one minus the cosine squared, so *any* non-zero directional agreement produces an improvement over naive. We have 0.57.

Both are implemented. Evaluating them is the next run.

Slide 13 — The model knows when it should not be trusted · 1 minute 20 seconds

On screen: three stat tiles, a wide statement box, two cards.

And this is the result we are proudest of, and it is independent of the accuracy ranking entirely.

We sample five futures from the same input and measure how much they disagree. The correlation between that disagreement and the error the model actually makes is 0.69.

In plain terms: when the five sampled futures disagree with each other, the prediction really is worse. The model knows when it is in trouble. Uncertainty also rises smoothly with how far ahead we predict — 0.00011 at one frame, 0.00055 at ten, roughly five times.

Now make the clinical argument. This is your strongest ground.

Why this matters more than the accuracy number: a system that is confidently wrong is unusable in an operating theatre. A system that can say "do not trust me here" is usable even when it is imperfect. And no method in our comparison provides this — not the naive baseline, not constant velocity, not a deterministic network. It is a direct consequence of the generative formulation.

Say the caveat yourself, before the panel raises it.

One caveat, stated carefully: this is a correlation between predicted spread and realised error. Some of it may reflect that busy moments are simply both harder and more variable. We say exactly that in the paper rather than claiming more than we measured.

Handover: "That is where our own numbers stand. Humair will close by putting them next to the published work."

Shehroz — questions likely to come to you

- **"So your model does not work?"** — It does not beat the baseline on Chamfer Distance, and we report that. It does learn motion direction at 0.57, and it does produce calibrated uncertainty at 0.69. Those are two measured positives and one measured negative, and we have localised the cause of the negative to magnitude.
- **"Why not just use constant velocity then?"** — For accuracy at one frame, today, you should. Constant velocity gives you no uncertainty, no multiple futures, and no mechanism to improve. Our point is that we now know exactly what to fix to pass it.
- **"Is 0.69 correlation actually good?"** — For an uncertainty estimate that was never trained against an error signal, yes. It is emergent from sampling, not fitted.
- **"How do you know your evaluation is not broken?"** — That is precisely what QT0 is for, and it is the first row on slide 9. The trajectory rebuild is verified against Module 4's saved windows before anything is predicted.
- **"Why Chamfer Distance?"** — It is correspondence-free, which matters because point identity is not guaranteed across frames. We also report Mean Point Error in the comparative study for the

> correspondence-aware view.

4. Humair Khan (22i-2623) — Slides 3, 4, 14, 15, 16

Opens the context, then closes the talk: FYP-1's reconstruction, the literature gap, the comparison with published work, future improvements and the summary. ≈ 5 minutes 20 seconds.

Slide 3 — FYP-1 built the reconstruction, FYP-2 adds the prediction · 50 seconds

On screen: the six-module pipeline strip, four stat tiles. Keep this fast — it is a recap, not a result.

Thank you, Ahmad. One minute on what already exists, so you know what Modules 5 and 6 are built on top of.

Trace the pipeline boxes left to right.

Modules 1 to 4 are complete and validated from FYP-1. Frames and masks, then 4D Gaussian reconstruction, then trajectory extraction, then self-supervised training windows. Modules 5 and 6, the two in red, are FYP-2 and are what the rest of this presentation covers.

The reconstruction reaches 30.10 dB PSNR and 0.904 SSIM from a single monocular camera, tracking roughly 58,000 Gaussians. From that we built 7,371 training windows with no human annotation at any point in the chain. That last part matters — nothing downstream of here required a surgeon to label anything.

Slide 4 — Four adjacent fields, none occupies the cell we need · 1 minute 10 seconds

On screen: a five-row table. The last row is highlighted. Work the columns, not the rows — the argument is that no existing row has all four properties.

This is the literature, and I want to be precise about the gap rather than claim nobody has done anything nearby, because that would not be true.

Four lines of work are adjacent to ours. Surgical 4D reconstruction — EndoNeRF, LerPlane, EndoGaussian, Deform3DGS — produces exactly the 3D geometry we want, but retrospectively. It does not predict the future.

Surgical video anticipation does predict the future — TPG-VAE and the recent SurgVista work — but in 2D pixel space, and conditioned on knowing the instrument's action in advance. There is no 3D geometry in the output.

Point cloud forecasting predicts 3D and does it well, but on rigid outdoor LiDAR scenes — PointRNN, MoNet, PCPNet. Rigid bodies and deforming soft tissue are very different problems.

And medical point diffusion — Xie et al. this year on liver motion — uses diffusion over 3D points on

exactly our kind of tissue, but estimates the *current* state from an X-ray projection, not a future one.

Now the last row and the last column.

Our claim is the bottom row, and it is a claim about the combination, not about any single column. Future 3D tissue geometry, from monocular video, with no annotation, and with a calibrated confidence estimate. Note the last column in particular — none of the four lines above us reports uncertainty at all.

Handover: "That is the gap. Ahmad will now take you through the two modules that go after it."

Slide 14 — How we compare with published work · 1 minute 20 seconds

On screen: left = reconstruction PSNR table plus a mint card; right = three cards on the prediction base papers.

Thank you, Shehroz. Dr Jamil asked us to identify base papers and compare against them properly, so we have two families.

Left side first.

For reconstruction, the base paper family is LerPlane, EndoGaussian and Deform3DGS. They report 29.46, 30.25 and 30.48 dB. We report 30.10.

State the caveat immediately — do not let them find it.

That comparison is indicative, not controlled, and we say so in the paper: those published figures use *stereo* supervision on different sequences. What it does establish is that our reconstruction sits within 0.4 dB of the stereo state of the art while using a single camera and no depth sensor — and that geometry is the input Module 5 learns to predict.

Right side.

For prediction, two base papers. PointRNN by Fan and Yang is the closest published method to what we are doing — predicting future point clouds from past ones, scored with Chamfer Distance — and we have implemented it faithfully as our primary learned baseline. And Luo and Hu's CVPR 2021 paper is the first diffusion model over 3D point clouds; our denoiser is a conditional, temporally grounded instance of that idea.

The full comparison is eleven methods on one protocol — five classical, five learned, and ours. Module 7 is implemented; the run is the next step.

Slide 15 — What comes next · 1 minute

On screen: five numbered rows. The order is the message.

Five things, and the order matters — the first two are the ones that could change the headline result, and both are already written.

One: evaluate the two fixes Shehroz described. Calibration needs no retraining at all, and on the mathematics he showed you, it should move accuracy below naive.

Two: retrain on the constant-velocity residual, which raises the model's floor from the naive baseline to one that is 41% stronger.

Three: hold out an entire video rather than frames from videos we also train on — that removes the optimism of selecting and reporting on the same material.

Four: complete the eleven-method comparison. Modules 7 and 8 are implemented; the runs remain.

Five: grow the dataset beyond ten short recordings. The overfitting we measured is entirely consistent with a corpus this size.

Slide 16 — Where Modules 5 and 6 stand · 1 minute

On screen: dark closing slide. Delivered / Not yet achieved, and the closing line.

To close, plainly.

Delivered: a complete prediction pipeline that needs no clinical labels. A six-test evaluation framework with an integrity gate. Calibrated uncertainty at 0.69. A comparative study across eleven methods, implemented. And a conference paper drafted in IEEE format.

Not yet achieved: beating the naive baseline on accuracy. We are not going to present that as anything other than what it is.

But we did not leave it unexplained. We localised the cause to displacement magnitude rather than direction, and we implemented two fixes that follow directly from that measurement. Evaluating them is the immediate next step.

Pause. Deliver the last line slowly — it is the sentence you want them to remember.

An honest negative result with an identified cause is worth more than an unexplained positive one.

Thank you. We are happy to take questions.

> **Humair — questions likely to come to you**
>
> - **"Your PSNR is lower than Deform3DGS. Isn't that worse?"** — It is 0.38 dB lower, and they use stereo input on different sequences. We state that on the slide rather than presenting it as a controlled win. The relevant point is that monocular reconstruction at this quality is good enough to serve as a prediction target.
> - **"Has nobody really predicted future surgical geometry?"** — People have predicted future surgical *pixels*, with instrument actions as a condition. People have predicted future 3D for rigid outdoor scenes. People have used point diffusion on tissue for the current state. The combination on slide 4's last row is what we have not found in the literature.
> - **"Where is the rendered output? Can we see a prediction?"** — Yes. We have side-by-side video of real frame, reconstruction, prediction and baseline, with magnified motion arrows, plus a forecast that runs twelve frames past the end of the recorded video. We can show it after the presentation.

- **"When will Modules 7 and 8 actually run?"** — Both are implemented and tested; they are queued against our Colab compute budget. They are items 4 on the next-steps slide for that reason.

5. Shared Question Routing

Agree this beforehand so you do not all start answering at once. Whoever is asked a question that belongs to someone else says one sentence and passes it: *"That is Shehroz's area — Shehroz?"*

If the question is about…	It goes to
The network, the diffusion formulation, why diffusion at all, training, overfitting, hyperparameters, the horizon conditioning, the residual target	**Ahmad**
Metrics, baselines, the quality tests, evaluation integrity, the comparison, the cosine finding, uncertainty, calibration, statistics	**Shehroz**
Reconstruction quality, rendering, the visual outputs, the forecast video, the literature, the base papers, the dataset, the runtime environment	**Humair**
Scope, timeline, division of work, what remains for FYP-2	**Ahmad** (then any member adds their part)

The one question to rehearse as a team. "Why should we pass a project that does not beat a two-line baseline?" Agree on a single answer and let one person give it. Suggested: *"Because we can tell you exactly why it does not, and the diagnosis is itself a result. We measured that the model learns motion direction at 0.57 and fails only on magnitude, we eliminated drift and dropped frames as explanations, and we implemented the two corrections that follow. We also produced a calibrated uncertainty estimate that none of the baselines can give you. We would rather bring you a measured negative with a cause than an unexplained positive."*